A conditional-span improvement and a resonance obstruction

Research note: fixed-target route, 27 October 2026

1 What a strict count comparison actually needs

For a centered variance-one law put $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $m = \mathbb{E}X_+$, $\ell = m/(pq)$, and $X = \ell(B + U - p)$, where $B = \mathbf{1}_{X>0}$ and $\mathbb{E}(U \mid B) = 0$. Write $s = \mathbb{E}U^2$ and $\lambda = ns$. Let $np + H/\ell = k + u$, with an interior integer count k and $|u| \le 1/2$. All variables are constructed from the same marginal.

For fixed target $c = C_{\mathrm{E}}$, define $a = \sqrt{n}\{F_n(H) - \Phi(H/\sqrt{n})\} - c\rho$. The exact count identity has left side

$$a + \rho_B(c - C_B),$$

where C_B is the actual matched-binomial score. If $a \ge 0$ and the external Bernoulli theorem gives $C_B \le c$, this is nonnegative. Thus a whole-count upper bound of the form

$$\sqrt{n}\pi_k \lambda B(\lambda) + \frac{\lambda}{n} I(\lambda)$$

is contradictory for $\lambda > 0$ whenever valid constants satisfy

$$B(\lambda) < 0, \qquad m_0 B(\lambda) + \frac{\max(0, I(\lambda))}{N} < 0, \quad \sqrt{n}\pi_k \ge m_0 > 0, \quad n \ge N. \tag{1}$$

The first sign gate is necessary before applying a lower mass bound. Zero noise is handled separately by the Bernoulli theorem.

Neither $D < 0.784$ nor a raw Fourier bound below 2.24 occurs in (1). Those are prescribed outputs of the retained initial acceptor. They supply later acquisition inputs through its particular dependency chain; they are not necessary scalar constants in every possible count proof. Conversely, obtaining a smaller D alone does not establish the conditional support, amplitude, marked-crossing and Gaussian-repair hypotheses needed to produce B, I. The example below shows why.

There is a useful exact test for the remaining marked parameter. On a fixed positive noise value, the retained whole-count formula has the form

$$B(\lambda) = z + \frac{A_\lambda}{z} - F_\lambda, \quad A_\lambda \ge 0, \quad 0 < z \le 1/U,$$

where $\eta = z\lambda \le 1$ on the noise band $[L, U]$. Write $F = F_\lambda - \max(0, I(\lambda))/(Nm_0)$ and $Z = 1/U$. Then (1) follows at this noise value precisely when $z + A_\lambda/z < F$ for an allowed z. For $A_\lambda > 0$ the minimum is

$$\min_{0<z\le Z} \left(z + \frac{A_\lambda}{z} \right) = \begin{cases} 2\sqrt{A_\lambda}, & Z^2 \ge A_\lambda, \\ Z + A_\lambda/Z, & Z^2 < A_\lambda. \end{cases} \tag{2}$$

This follows by differentiating, or completing $(\sqrt{z} - \sqrt{A_\lambda/z})^2$ and checking the boundary. If $A_\lambda = 0$, feasibility is exactly $F > 0$; the infimum zero is not attained. In the interior case the exact rational gate is $F > 0$ and $F^2 > 4A_\lambda$. Equality does not suffice. For a full closed noise band a common z must still satisfy both endpoint gates before using the existing convexity proof; optimizing z separately at the endpoints does not establish that common witness. In particular a zero fractional-part gain makes this whole-count formula nonnegative for every $z > 0$, even if the mark cost vanishes. It cannot be repaired simply by shrinking the weighted resource.

2 The harmonic argument works up to span one

For a real number x, write $\{x\} = x - \lfloor x \rfloor$.

Lemma 1 (Extended conditional-span domain)**.** *Let $Z_1, \dots, Z_j$ be independent real variables, each supported in an interval of length at most b, where $0 < b < 1$. Put $W = \sum_i Z_i$ and $t = \sum_i \operatorname{Var}(Z_i)$. Suppose $K > 0$ satisfies*

$$1 - \cos(2\pi x) \le Kx \quad (0 \le x \le 1).$$

For every $w \in \mathbb{R}$ and every $0 < r \le 2\sin^2(\pi b)/b^2$,

$$\mathbb{E}\{w - W\} \ge K^{-1}(1 - e^{-rt}). \tag{3}$$

In particular $K = 8$ is valid without a finite scalar certificate. The inherited value $K = 23/5$, or its accepted improvement, may be retained when its separate scalar majorization has been proved.

Proof. Let Z_i' be an independent copy. Since $|Z_i - Z_i'| \le b$ and sine is concave on $[0, \pi]$, the chord from 0 to πb gives

$$\sin^2\big(\pi(Z_i - Z_i')\big) \ge \frac{\sin^2(\pi b)}{b^2}(Z_i - Z_i')^2.$$

Consequently, writing $v_i = \operatorname{Var}(Z_i)$,

$$|\mathbb{E}e^{2\pi i Z_i}|^2 = 1 - 2\mathbb{E}\sin^2\big(\pi(Z_i - Z_i')\big) \le 1 - \frac{4\sin^2(\pi b)}{b^2}v_i \le e^{-2rv_i}.$$

The intermediate right side is nonnegative because it bounds the nonnegative left side. Taking square roots and multiplying, using independence, gives $|\mathbb{E}e^{2\pi i W}| \le e^{-rt}$. Periodicity then gives

$$K\mathbb{E}\{w - W\} \ge 1 - \Re\mathbb{E}e^{2\pi i(w-W)} \ge 1 - |\mathbb{E}e^{2\pi i W}| \ge 1 - e^{-rt}.$$

No centering or continuous distribution assumption was used. To verify $K = 8$, the function $y - 1 + \cos y$ is nondecreasing for $y \ge 0$, since its derivative is $1 - \sin y \ge 0$. Thus $1 - \cos(2\pi x) \le 2\pi x < 8x$ for $x > 0$; $\pi < 4$ also follows directly from $\pi = 4\int_0^1 (1 + y^2)^{-1}\, dy < 4$. Equality at $x = 0$ is harmless. □

The earlier count proof restricts $b < 1/2$ at this step. The same proof in fact gives Lemma 1 throughout $b < 1$. For example, $b = 3/4$ permits the exact rate $r = 16/9$. This extends only the harmonic component. It does not extend other denominator, binomial, support or noise gates. In the retained whole-count notation a sufficient acquired span is $b = A + d_0$; it must be a bound on each actual conditional residual, not on its variance.

Proposition 2 (The boundary is a real obstruction)**.** *No strictly positive uniform rate in* (3) *is possible for all independent positive-variance summands of support span at most one. Even a uniformly positive rate as $b \uparrow 1$ is impossible.*

Proof. For $0 < \varepsilon < 1$, let Z equal $\varepsilon - 1$ with probability ε, and ε otherwise. It is centered, has positive variance $\varepsilon(1 - \varepsilon)$ and span one. For j independent copies, $W = j\varepsilon - M$ with integer-valued M, so $\{j\varepsilon - W\} = 0$ identically. The right side of (3) is positive for every $r > 0$ and finite $K > 0$.

For the boundary limit, the centered variable taking values $\pm b/2$ with equal probabilities has variance $b^2/4$ and characteristic modulus $|\cos(\pi b)|$. As $b \uparrow 1$, this modulus tends to one while the variance tends to $1/4$. A positive uniform exponential decay rate would contradict these limits. In fact $1 - |\cos(\pi b)|$ vanishes quadratically in $1 - b$, consistently with the rate in Lemma 1. □

3 An actual iid law with arbitrarily small moment resources

The obstruction can be realized by the actual conditional laws of a centered, variance-one iid marginal with a large central gap. It is not merely an optimizer of a moment relaxation.

For rational $\varepsilon, t > 0$, set

$$q = \frac{1+\varepsilon-\varepsilon^2}{1+t^2}, \quad p = 1-q, \quad r = tq, \qquad 0 < \varepsilon < p < 1. \tag{4}$$

Let Y take the values $\varepsilon - 1, \varepsilon, 1$ with respective probabilities $q\varepsilon, q(1-\varepsilon), p$, and put $X = (Y-p)/r$. Direct calculation gives

$$\mathbb{E}Y = p, \quad \mathrm{Var}(Y) = q\{p + \varepsilon(1-\varepsilon)\} = r^2.$$

Therefore X is exactly centered and standardized. The first two atoms are negative, and the third is positive. In the actual sign split, $\ell = 1/r$, $U_1 = 0$, and

$$U_0 = \begin{cases} \varepsilon - 1, & \text{probability } \varepsilon, \\ \varepsilon, & \text{probability } 1-\varepsilon. \end{cases}$$

Thus the actual conditional amplitude is $1-\varepsilon$, its span is one, the normalized central-gap increment is $c_{\mathrm{gap}} = 1-\varepsilon$, and $\lambda = nq\varepsilon(1-\varepsilon)$. In particular the central gap is wide even when the harmonic estimate fails completely.

For a fixed interior count k, the actual conditional sum has the law

$$V_k = (n-k)\varepsilon - \mathrm{Bin}(n-k, \varepsilon).$$

With $w = (n-k)\varepsilon$, its fractional-part gain is zero. For the four examples checked below, $n = 42$, $k = 17$ and $u = \eta = w/2$ give a legal phase $|u| \le 1/2$, $0 < \eta \le 1$, and $w = u + \eta$, exactly as in the marked-crossing argument. Its own-count variance is $(n-k)\varepsilon(1-\varepsilon) > 0$.

The physical defect is computed from the actual gaps, not from the two-point skeleton:

$$\rho = \mathbb{E}|X|^3, \quad \gamma = \mathbb{E}X^3, \quad J = \sum_{x_i < x_{i+1}} \left(\sum_{x_j \ge x_{i+1}} p_j x_j \right) (x_{i+1} - x_i)^2, \qquad D = (3+\sqrt{10})\rho - \gamma - 3J.$$

The exact checker uses rational moments and the directed enclosure

$$3.162277660168379331 < \sqrt{10} < 3.162277660168379332,$$

verified by squaring positive rationals. Its first law, with $\varepsilon = 1/100$, $t = 21/25$, is explicitly

x	$-198687/70693$	$-169661/212079$	$25/21$
$\mathbb{P}(X = x)$	$10099/1705600$	$999801/1705600$	$6957/17056.$

It has $|X| < 2.811$, $1.11975 < \rho < 1.11976$, and $0.67024 < D < 0.670241$. The four fixed examples satisfy:

ε	t	strict upper bound for D	strict upper bound for λ
10^{-2}	$21/25$	0.670241	0.246199
10^{-3}	$21/25$	0.070624	0.024625
10^{-6}	$21/25$	0.000738	0.000025
10^{-9}	$849/1000$	0.0000001	0.0000001

Every example satisfies its own recorded v6 moment-cell defect bound, weighted nD bound, and strict actual-gap criterion, as well as $D < 0.784$. This is a fresh exact comparison with the pinned table, not a replay of the Fourier calculation that originally produced that table.

Corollary 3 (No moment-only amplitude implication)**.** *For every $\delta > 0$ there is an actual centered variance-one finite law supported in $[-96/25, 96/25]$, with positive conditional noise, $D < \delta$ and $\lambda < \delta$ at $n = 42$, whose negative conditional residual span is exactly one and whose fractional-part gain vanishes at a legal marked phase. The normalized central-gap increment can be arbitrarily close to one.*

Proof. At $\varepsilon = 0$ the positive-weight part of (4) is the standardized two-point law at $-t, 1/t$, with positive probability $p = t^2/(1+t^2)$. Its defect is

$$D_B = \frac{(6+2\sqrt{10})(p-p_*)^2}{\sqrt{p(1-p)}}, \qquad p_* = (4-\sqrt{10})/2.$$

This follows by substituting ρ_B, γ_B and $J_B = t + 1/t$. Choose positive rational t arbitrarily near $\sqrt{p_*/(1-p_*)}$. Then D_B is arbitrarily small. For this fixed t, let positive rational ε tend to zero. All moments are continuous. The extra outer-gap weight tends to zero, so the displayed finite gap formula gives $D \to D_B$ as well; $\lambda \to 0$ while the conditional span remains one. At the limiting value of t, the outer atom tends to $-(1+2t^2)/t$. Here $0.84 < t < 0.85$, so its magnitude is less than $2(0.85) + 1/0.84 < 3$. The support is therefore within $[-96/25, 96/25]$ in a neighborhood. The sign, phase and positive-noise conditions also hold for all sufficiently small ε. □

These laws are not asserted to satisfy a maximizing contact inequality, nonnegative fixed-target excess, or all new signed-acquisition outputs. They do not refute the sharp conjecture or the retained local theorem. They refute the proposed implication from small moment resources and an actual central gap alone to a positive fractional-part gain or a small conditional amplitude. Contact-based confinement or a quantitatively proved replacement for harmonic nonresonance remains an essential missing edge in that shortcut.